

Q1.

(a) Prove that $\frac{\sinh \theta}{1 + \cosh \theta} + \frac{1 + \cosh \theta}{\sinh \theta} = 2 \coth \theta$

Explicitly state any hyperbolic identities that you use within your proof.

(4)

(b) Solve $\frac{\sinh \theta}{1 + \cosh \theta} + \frac{1 + \cosh \theta}{\sinh \theta} = 4$ giving your answer in an exact form.

(2)

(Total 6 marks)

Q2.

(a) Sketch the curve $y = \cosh x$.

(1)

(b) Solve the equation

$$6 \cosh^2 x - 7 \cosh x - 5 = 0$$

giving your answers in logarithmic form.

(6)

(Total 7 marks)

Q3.

(a) Use the definitions of $\cosh \theta$ and $\sinh \theta$ in terms of e^θ to show that

$$\cosh x \cosh y - \sinh x \sinh y = \cosh(x - y)$$

(4)

(b) It is given that x satisfies the equation

$$\cosh(x - \ln 2) = \sinh x$$

(i) Show that $\tanh x = \frac{5}{7}$.

(4)

(ii) Express x in the form $\frac{1}{2} \ln a$.

(2)

(Total 10 marks)

Q4.

(a) Use the definitions $\cosh x = \frac{1}{2}(e^x + e^{-x})$ and $\sinh x = \frac{1}{2}(e^x - e^{-x})$ to show that

(3)

- (b) (i) Express

$$5 \cosh^2 x + 3 \sinh^2 x$$

in terms of $\cosh x$.

(1)

- (ii) Sketch the curve $y = \cosh x$.

(1)

- (iii) Hence solve the equation

$$5 \cosh^2 x + 3 \sinh^2 x = 9.5$$

giving your answers in logarithmic form.

(4)

(Total 9 marks)

Q5.

- (a) Show that

$$9 \sinh x - \cosh x = 4e^x - 5e^{-x}$$

(2)

- (b) Given that

$$9 \sinh x - \cosh x = 8$$

find the exact value of $\tanh x$.

(7)

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(Total 9 marks)

Q6.

- (a) Given that

$$4 \cosh^2 x = 7 \sinh x + 1$$

find the two possible values of $\sinh x$.

(4)

- (b) Hence obtain the two possible values of x , giving your answers in the form $\ln p$.

(3)

(Total 7 marks)